

AZ-104

Tag 2

Administer Azure Resources

Guten Morgen!



AZ-104 Agenda

01: Administer Identity

02: Administer Governance and Compliance

03: Administer Azure Resources

04: Administer Virtual Networking

05: Administer Intersite Connectivity

06: Administer Network Traffic Management

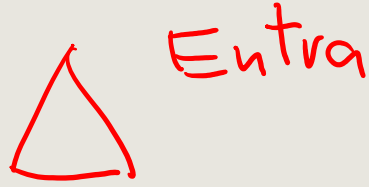
07: Administer Azure Storage

08: Administer Azure Virtual Machines

09: Administer PaaS Compute Options

10: Administer Data Protection

11: Administer Monitoring



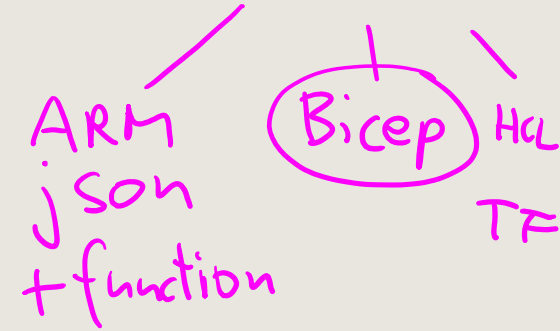
Tool

PS

Portal

Azure CLI

Templates



Learning Objectives - Azure Resources

- Configure Azure Resources with Tools
 - Microsoft Copilot in Azure
 - Manage services with the Azure portal
 - Introduction to PowerShell
 - Introduction to Bash
- Deploy Azure resources with templates
- Lab 03b - Manage Azure resources by Using ARM Templates

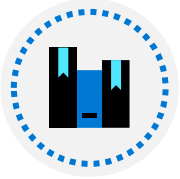
Configure Azure Resources with the Azure Portal, PowerShell, and the CLI



Learning Objective – Configure Azure Resources with tools

- Microsoft Copilot for Azure
- Other Administration tools
 - Azure Portal
 - Azure Cloud Shell
 - Azure PowerShell and CLI
- Demonstration – Azure Tools
- Learning Recap

Review Azure Resource Terminology



A resource is simply a single service instance in Azure

Terraform (IBM)
HCL
plugin
AWS
AzureRM
:



A resource group is a logical grouping of resources

Bicep Azure Only



An Azure Resource Manager template is a JSON file that allows you to declaratively describe a set of resources



A declarative syntax is what a template uses to state what you intend to create



A **resource provider** is service that supplies the resources you can deploy and manage through Resource Manager

idempotent
Semper idem!

Review Resource Manager Benefits

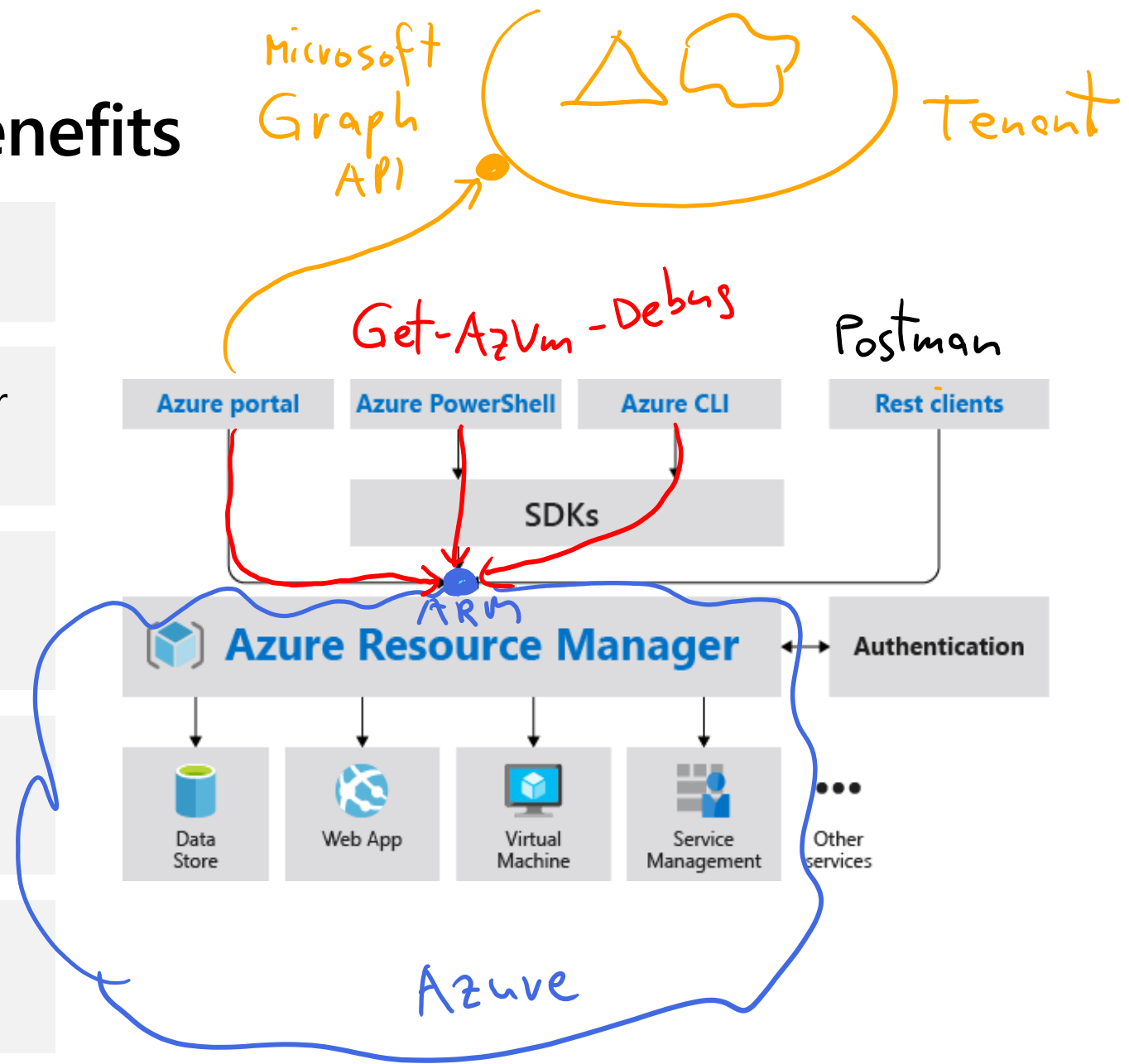
Provides a consistent management layer

Enables you to work with the resources in your solution as a group

Deploy, update, or delete in a single, coordinated operation

Provides security, auditing, and tagging features

Choose the tools and APIs that work best for you



Use Azure Cloud Shell

Interactive, browser-accessible shell

Offers either Bash or PowerShell

Is temporary and provided on a per-session, per-user basis

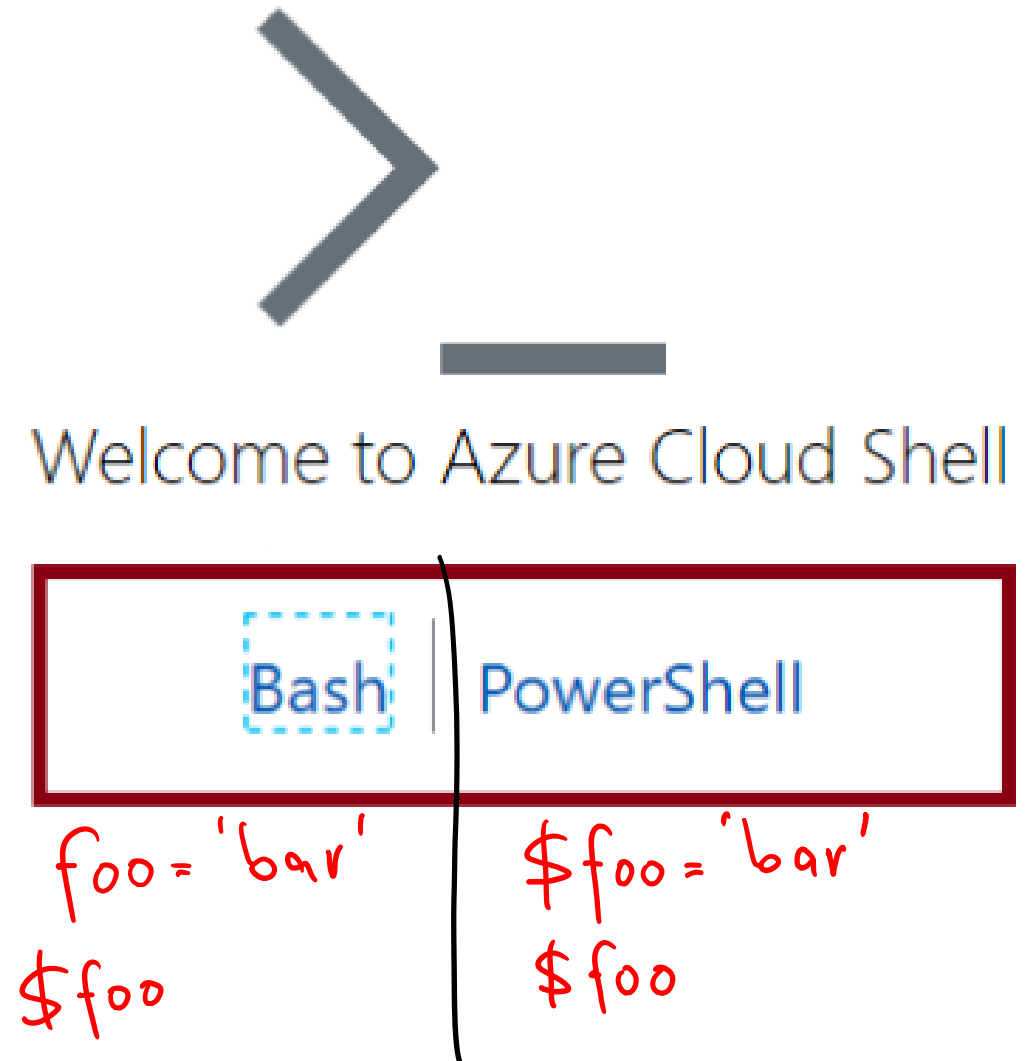
Requires a resource group, storage account, and Azure File share

Authenticates automatically

Integrated graphical text editor

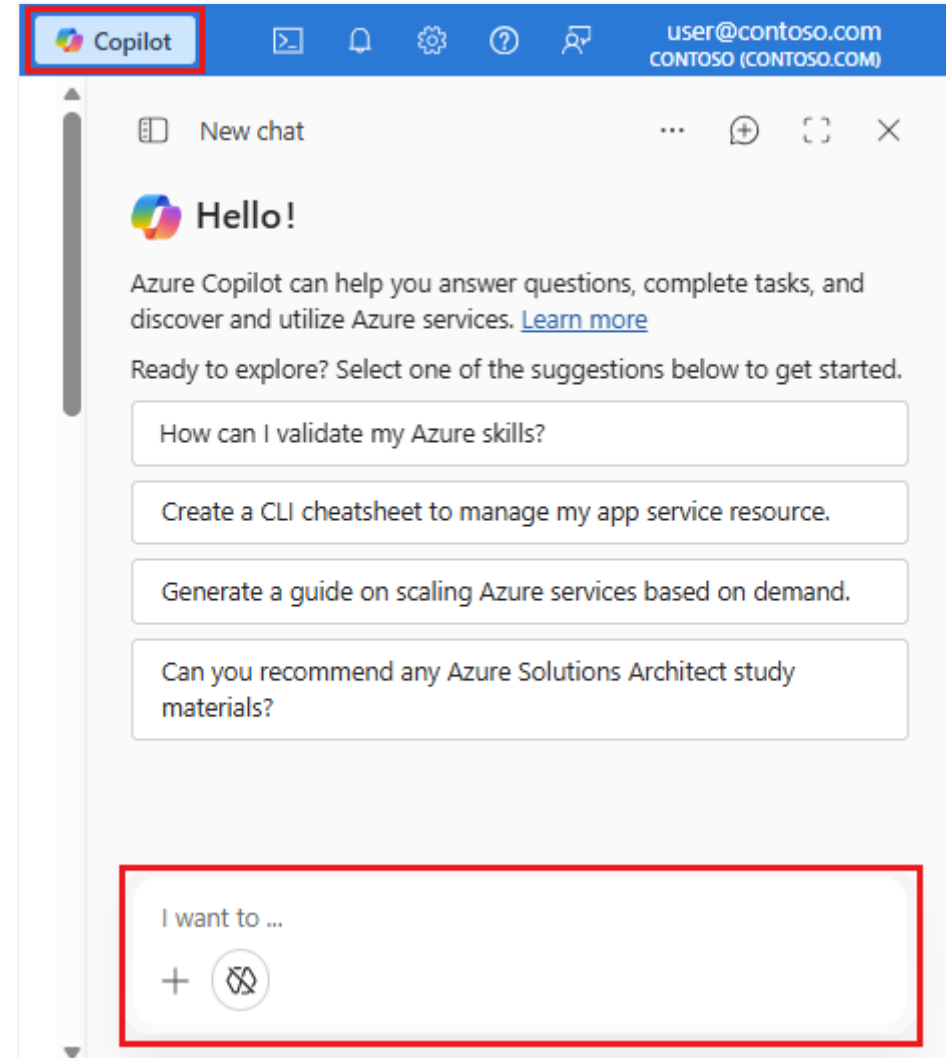
Is assigned one machine per user account

Times out after 20 minutes



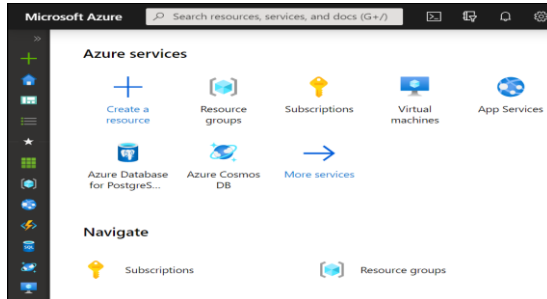
Microsoft Copilot for Azure

- Access in the Azure portal, through the Azure mobile app, or through AI Shell
- Helps you design, operate, optimize, and troubleshoot your Azure apps and infrastructure
- Performs many basic tasks
 - Deploy and manage virtual machines
 - Design, troubleshoot, and secure networks
 - Generate PowerShell and Azure CLI scripts
 - Analyze, estimate, and optimize costs
 - Discover and deploy workload templates



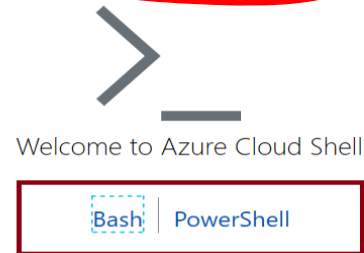
Other Administrator Tools

Azure Portal



- View and manage resources
- Visual interface
- Unified hub – training and documentation
- Personalize your experience
- Mobile app
- Access the Cloud Shell
- One-off creation scenarios

Azure Cloud Shell



- Interactive and browser-accessible with file storage
- Offers Bash or PowerShell
- Authenticates automatically
- Provided on a per-session and per-user basis
- Temporary - times out after 20 minutes

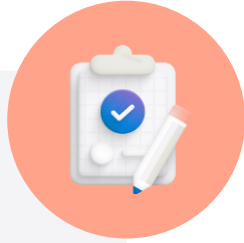
Azure PowerShell and CLI

```
az vm restart -g  
MyResourceGroup -n MyVm
```

Get-AzVM

- Command line programs
- Interactive and scripting modes
- Cross-platform
- Good for repeatable deployments
- Familiar coding experience

Learning Recap – Azure Resources with Tools



Check your
knowledge
questions and
additional
study

Discussion Questions

Name at least three tools you can use to manage Azure resources. In what situations would you use each tool?

Reference modules

- [Describe features and tools for managing and deploying Azure resources](#)
- [Introduction to Azure Cloud Shell](#)
- [Introduction to Microsoft Copilot in Azure](#)

Deploy Azure resources with templates



Learning Objective – Deploy Azure resources using templates

- Explore the JSON Template Schema
- Explore the JSON Template Parameters
- Consider Azure Bicep Files
- Demonstration – QuickStart Templates
- Learning Recap

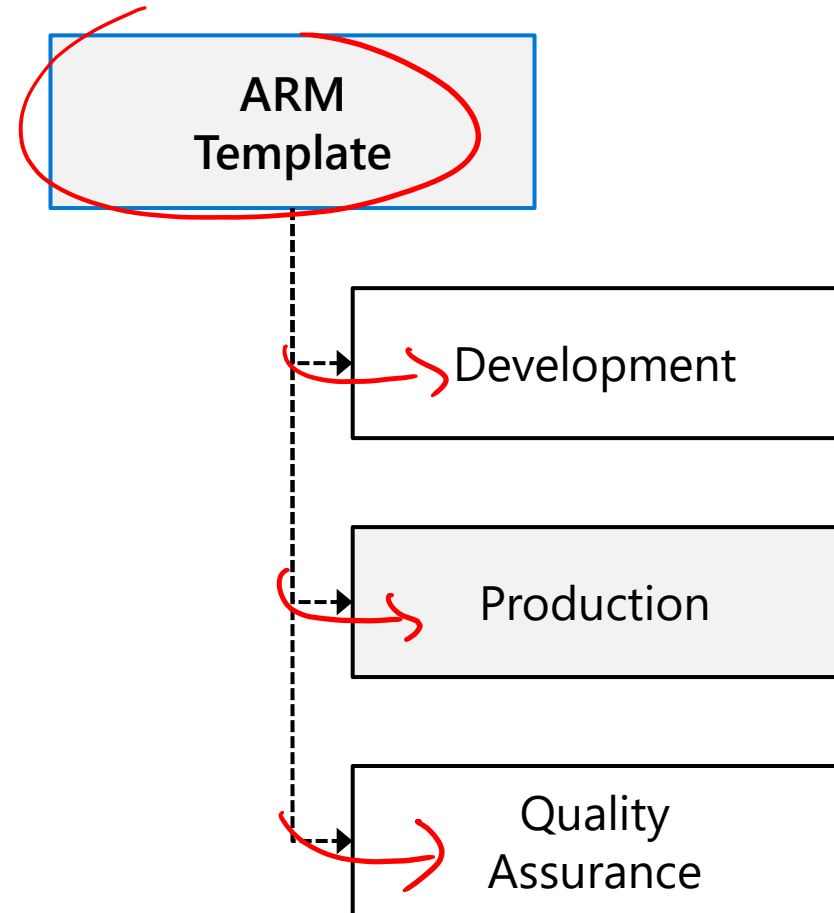
Exam objectives

Automate deployment of resources by using ARM templates or Bicep files

- Interpret an ARM template or a Bicep file
- Modify an existing ARM template
- Modify an existing Bicep file
- Deploy resources by using an ARM template or a Bicep file
- Export a deployment as an ARM template or convert an ARM template to a Bicep file

Review ARM Template Advantages

- Improves consistency and promotes reuse
- Reduce manual, error prone, and repetitive tasks
- Express complex deployments
- Express requirements through code
- Provides validation tasks
- Modular and can be linked
- Simplifies orchestration



Explore the JSON Template Schema

- Defines all the Resource manager resources in a deployment
- Written in JSON
- A collection of key-value pairs
- Each key is a string
- Each value can be a string, number, Boolean expression, list of values, object

```
{  
  "$schema":  
    "http://schema.management.  
    azure.com/schemas/2019-04-  
    01/deploymentTemplate.json#",  
  "contentVersion": "",  
  "parameters": {},  
  "variables": {},  
  "functions": [],  
  "resources": [],  
  "outputs": {}  
}
```

Explore the JSON Template Parameters

- Specifies which values are configurable when the template runs
- This example has two parameters: one for a VM's username (adminUsername), and one for its password (adminPassword)

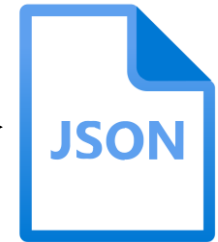
```
"parameters": {  
  "adminUsername": {  
    "type": "string",  
    "metadata": {  
      "description": "Username for the VM."  
    }  
  },  
  "adminPassword": {  
    "type": "securestring",  
    "metadata": {  
      "description": "Password for the VM."  
    }  
  }  
}
```


Consider Azure Bicep Files

- Simpler syntax for writing templates
- Smaller module files you can reference from a main template
- Automatically detect dependencies between your resources
- Visual Studio Code extension with validation and IntelliSense

Bicep file

```
resource storageAccount
'Microsoft.Storage/storageAccounts@
2021-01-01' = {
  name: storageAccountName
  location: location
  tags: {
    displayName: storageAccountName
  }
  kind: 'StorageV2'
  sku: {
    name: 'Standard_LRS'
  }
}
```

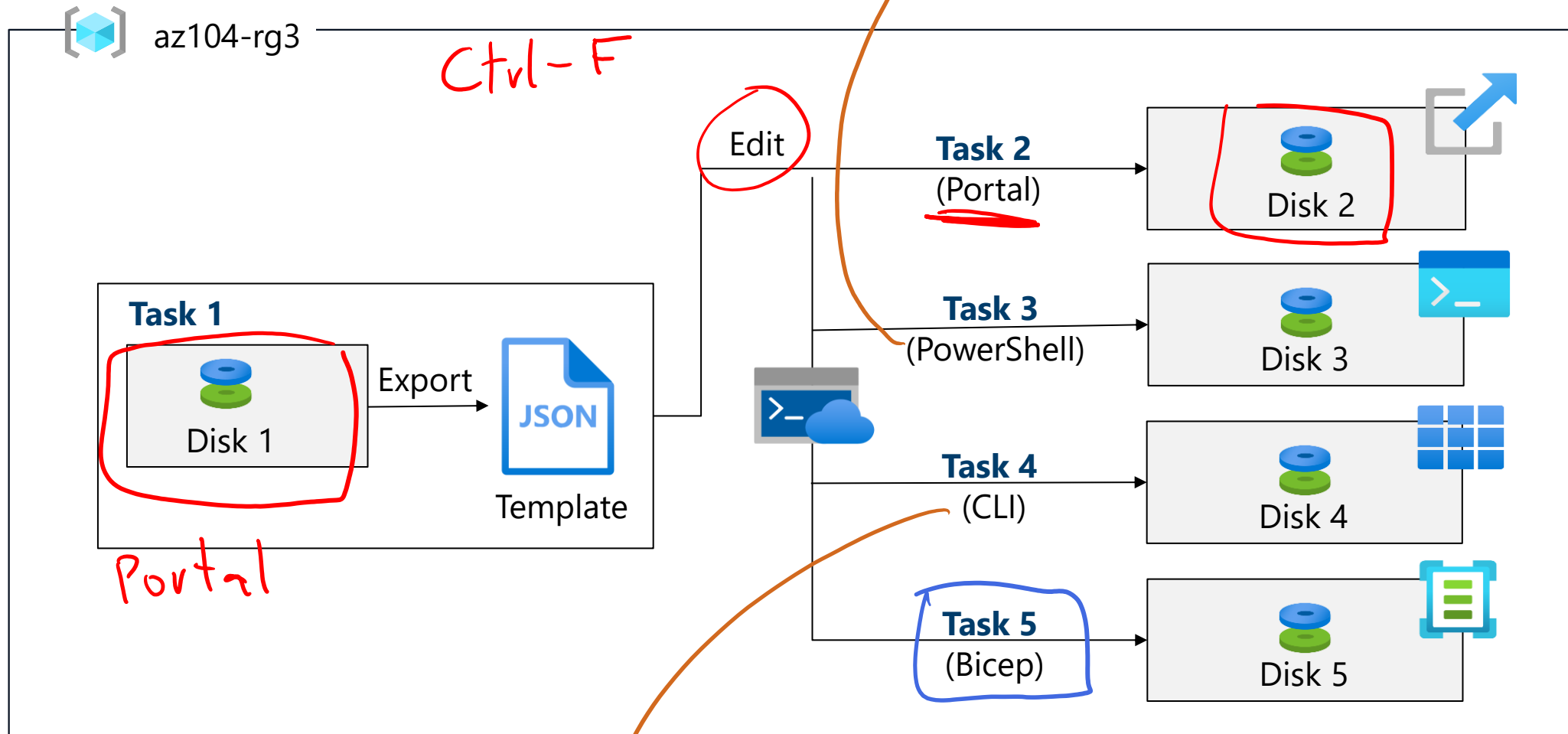


Lab - Manage Azure resources by Using ARM Templates



Lab 03 – Architecture diagram

New - Az Resource Group Deployment



End of presentation

